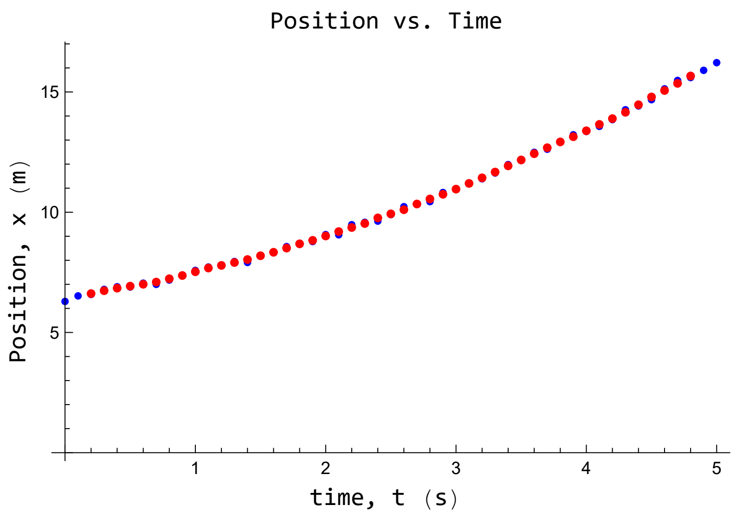
Problem 3.

```
In[ ]:= t = {0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7,
    1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3.0, 3.1, 3.2, 3.3, 3.4,
    3.5, 3.6, 3.7, 3.8, 3.9, 4.0, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 5.0};
x = {6.29, 6.52, 6.58, 6.79, 6.9, 6.89, 7.05, 7.00, 7.18, 7.36, 7.58, 7.72, 7.78, 7.95, 7.91,
    8.18, 8.34, 8.57, 8.67, 8.78, 9.07, 9.06, 9.48, 9.58, 9.63, 9.9, 10.23, 10.32,
    10.44, 10.82, 10.95, 11.2, 11.39, 11.63, 11.98, 12.15, 12.49, 12.62, 12.93, 13.22,
    13.36, 13.57, 13.85, 14.26, 14.42, 14.68, 15.13, 15.48, 15.6, 15.9, 16.22};
data = Transpose[{t, x}];
smoothedData = MovingAverage[data, 5];
Dataplot = ListPlot[data, PlotStyle → Blue];
Labeled[Show[Dataplot, ListPlot[smoothedData, PlotStyle → Red]],
 {"Position vs. Time", "time, t (s)", "Position, x (m)"},
 {Top, Bottom, Left}, RotateLabel → True]
Out[ ]=
```



In[ ]:=
```
v = Table[(x[[i + 2]] - x[[i]])/(2 dt), {i, 1, Length[x] - 2}];
dt = 0.1;
t2 = Table[t[[i]], {i, 2, Length[x] - 1}];
vdata = Thread[{t2, v}];
vfit = LinearModelFit[vdata, V, V];
vfitsmooth = LinearModelFit[vdatasmooth, V, V];
vfit["ParameterTable"]
vfitplot = Plot[vfit[X], {X, Min[t], Max[t]}, PlotStyle → Black];
vplot = ListPlot[vdata, PlotStyle → Green];

xsmooth = MovingAverage[x, 5];
tsmooth = MovingAverage[t, 5];
vsmooth = Table[(xsmooth[[i + 2]] - xsmooth[[i]])/(2 dt), {i, 1, Length[xsmooth] - 2}];
tsmooth = Table[tsmooth[[i]], {i, 2, Length[xsmooth] - 1}];
vdatasmooth = Thread[{tsmooth, vsmooth}];
vfitsmooth = LinearModelFit[vdatasmooth, V, V];
vfitsmooth["ParameterTable"]
vfitplotsmooth = Plot[vfitsmooth[X], {X, Min[tsmooth], Max[tsmooth]}, PlotStyle → Red];
vplotsmooth = ListPlot[vdatasmooth, PlotStyle → Blue];

Labeled[Show[vfitplot, vplot, vfitplotsmooth, vplotsmooth ],
 {"Velocity vs. Time", "time, t (s)", "velocity, v (m/s)"},
 {Top, Bottom, Left}, RotateLabel → True]

vds = MovingAverage[v, 5];
tds = MovingAverage[t2, 5];
dt = 0.1;
vdatads = Thread[{tds, vds}];
vfitds = LinearModelFit[vdatads, V, V];
vfitds["ParameterTable"]
vfitplotds = Plot[vfitds[X], {X, Min[tds], Max[tds]}, PlotStyle → Red];
Labeled[vplotds = ListPlot[vdatads, PlotStyle → Blue],
  {"Velocity vs. Time", "time, t (s)", "velocity, v (m/s)"},
  {Top, Bottom, Left}, RotateLabel → True];
Labeled[Show[vfitplot, vplot, vfitplotds, vplotds],
 {"Velocity vs. Time", "time, t (s)", "velocity, v (m/s)"},
 {Top, Bottom, Left}, RotateLabel → True]
```

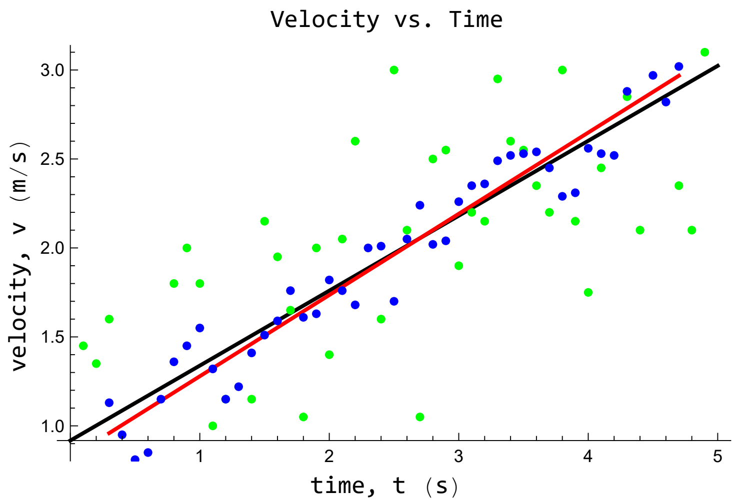
Out[ ]=

	Estimate	Standard Error	t-Statistic	P-Value
1	0.917347	0.164819	5.56579	1.21457×10^{-6}
V	0.421224	0.0573826	7.34064	2.49296×10^{-9}

Out[∘]=

	Estimate	Standard Error	t-Statistic	P-Value
1	0.822218	0.0515139	15.9611	1.11294×10^{-19}
V	0.456535	0.0182854	24.9672	3.20886×10^{-27}

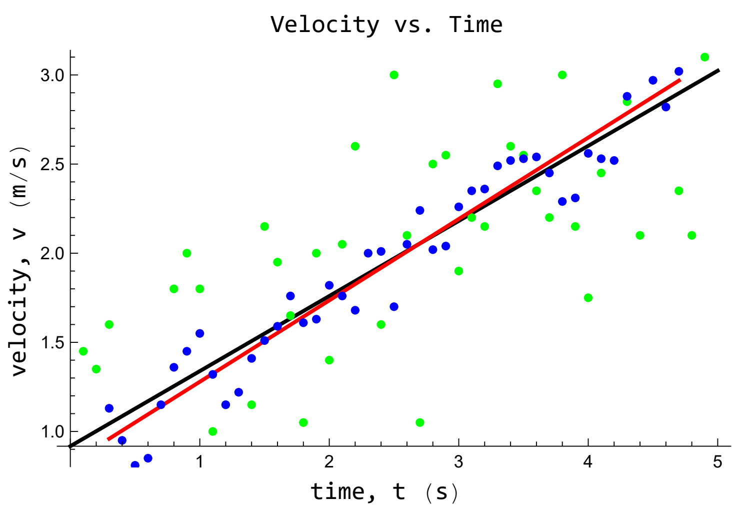
Out[∘]=



Out[∘]=

	Estimate	Standard Error	t-Statistic	P-Value
1	0.822218	0.0515139	15.9611	1.11294×10^{-19}
V	0.456535	0.0182854	24.9672	3.20886×10^{-27}

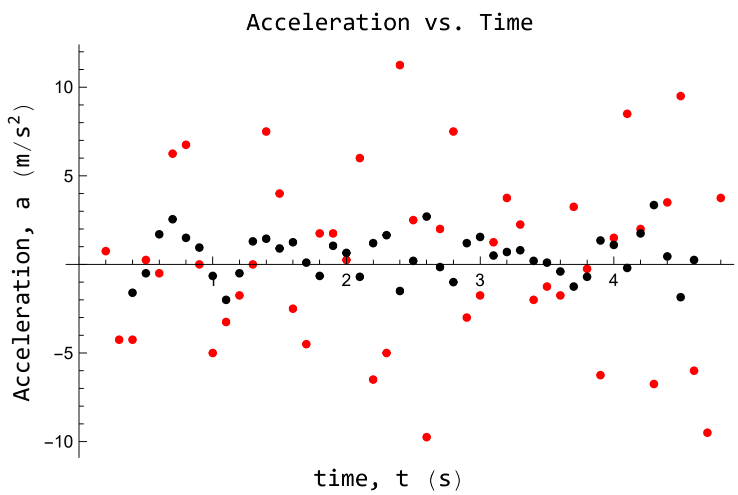
Out[∘]=



In[*]:=
```
a = Table[(v⟦i + 2⟧ - v⟦i⟧)/(2 dt), {i, 1, Length[v] - 2}];
t3 = Table[t2⟦i⟧, {i, 2, Length[v] - 1}];
adata = Thread[{t3, a}];
aplot = ListPlot[adata, PlotStyle → Red];

as = Table[(vsmooth⟦i + 2⟧ - vsmooth⟦i⟧)/(2 dt), {i, 1, Length[vsmooth] - 2}];
ts = Table[tsmooth⟦i⟧, {i, 2, Length[vsmooth] - 1}];
adatas = Thread[{ts, as}];
Labeled[Show[aplot, aplots = ListPlot[adatas, PlotStyle → Black]],
 {"Acceleration vs. Time", "time, t (s)", "Acceleration, a (m/s^2)"},
 {Top, Bottom, Left}, RotateLabel → True]
Print[" Mean Accelaration: ", Mean[a],
  " | STD: ", StandardDeviation[a] ,
 " | Smoothed Mean Accelaration: ", Mean[as],
  " | Smoothed STD: ", StandardDeviation[as]]
```

Out[*]=



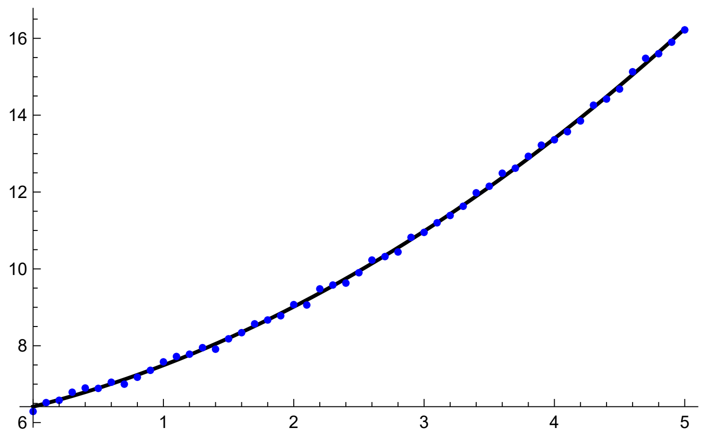

```
Mean Accelaration: 0.255319 | STD: 4.91545
 | Smoothed Mean Accelaration: 0.437209 | Smoothed STD: 1.23168
```

In[◦]:=
```
xfit = NonlinearModelFit[data, a0 + a1 X + 1/2 a2 X^2, {a0, a1, a2}, X];
xfit["ParameterTable"]
xfitplot = Plot[xfit[X], {X, Min[t], Max[t]}, PlotStyle → Black];
Show[xfitplot, Dataplot]
```

Out[◦]=

	Estimate	Standard Error	t-Statistic	P-Value
a0	6.41418	0.0302922	211.743	5.74196×10^{-73}
a1	0.855848	0.0280186	30.5457	4.15741×10^{-33}
a2	0.444292	0.0108384	40.9923	5.14572×10^{-39}

Out[◦]=

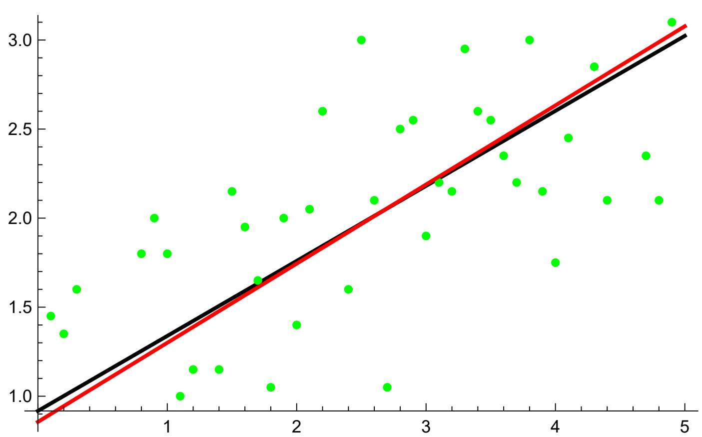



```
vfit = LinearModelFit[vdata, V, V];
vfit["ParameterTable"]
vfitplot = Plot[vfit[X], {X, Min[t], Max[t]}, PlotStyle → Black];
dvplot = Plot[(D[xfit[X], X] /. X → T), {T, Min[t], Max[t]}, PlotStyle → Red];
Show[vfitplot, vplot, dvplot]
```

Out[◦]=

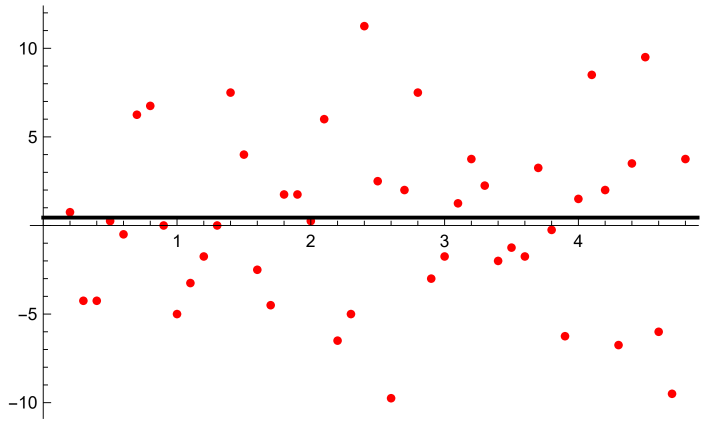
	Estimate	Standard Error	t-Statistic	P-Value
1	0.917347	0.164819	5.56579	1.21457×10^{-6}
V	0.421224	0.0573826	7.34064	2.49296×10^{-9}

Out[◦]=



In[◦]:=
```
daplot = Plot[D[xfit[X], {X, 2}] /. X → T, {T, Min[t], Max[t]}, PlotStyle → Black];
Show[aplot, daplot]
```

Out[◦]=

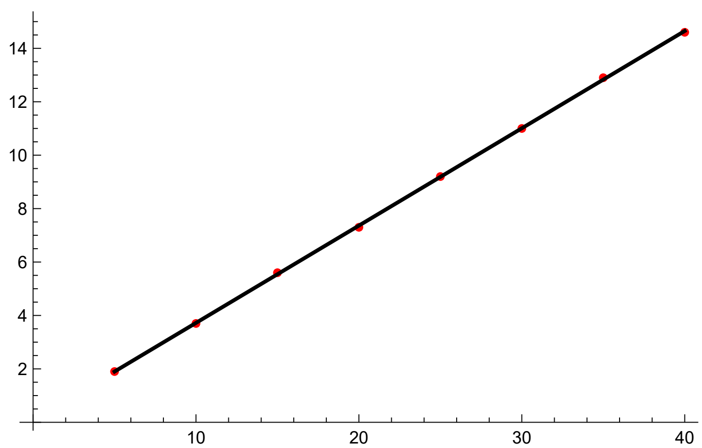


Problem 4.

In[58]:=
```
x = {5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0, 40.0};
F = {1.9, 3.7, 5.6, 7.3, 9.2, 11.0, 12.9, 14.6};
data = Thread[{x, F}];
lmf = LinearModelFit[data, X, X];
k = lmf["BestFitParameters"][[2]];
Print["Spring Constant k = ", k, " N/cm"];
dataplot = ListPlot[data, PlotStyle → Red];
fitplot = Plot[lmf[X], {X, Min[x], Max[x]}, PlotStyle → Black];
Show[dataplot, fitplot]
```

```
Spring Constant k = 0.364286 N/cm
```

Out[66]=



In[67]:=
```
rectangles = Table[x⟦i⟧ (F⟦i + 1⟧ - F⟦i⟧), {i, 1, Length[F] - 1}];
(* Create a rectangular approximation to the integral*)
intrect = Total[rectangles]
```

Out[68]=
```
253.
```

In[77]:=
```
trapezoids = Table[(x⟦i⟧ + x⟦i + 1⟧) / 2 (F⟦i + 1⟧ - F⟦i⟧), {i, 1, Length[F] - 1}];
(* Create a trapezoidal approximation to the integral*)
inttrap = Total[trapezoids]
```

Out[78]=
```
284.75
```

In[98]:= $F = \frac{1}{2} k * (\text{Max}[x])^2$

Out[98]=
```
291.429
```

In[99]:=
```
int = Integrate[lmf[X], {X, Min[x], Max[x]}]
```

Out[99]=
```
289.625
```

```
(* Comparing to results from the known function *)
```

In[100]:=
```
Print[Column[{"Percent Difference:",
   Row[{"Rectangular Integration: ", Abs[intrect - F] / F * 100, "%"}],
   Row[{"Trapezoidal Integration: ", Abs[inttrap - F] / F * 100, "%"}],
   Row[{"Linear Model Fit: ", Abs[int - F] / F * 100, "%"}]}, Spacings → 1]]
```

```
Percent Difference:

Rectangular Integration: 13.1863%

Trapezoidal Integration: 2.29167%

Linear Model Fit: 0.618873%
```